

04

Electric Vehicle
Codec Technologies India
Internship 2026

AI Driving Efficiency Optimizer

scikit-learn Wh/km regression, driver-style classifier and counterfactual tip engine on a synthetic fleet

AT A GLANCE

2,000 / 41,259

Synthetic trips / km

0.975

GradientBoosting test R2

5.04 Wh/km (3.4 %)

Test MAE

95.0 %

Style classifier accuracy

hvac_kw 0.577

Top feature (perm. importance)

19.9 %

Combined tip saving, aggressive

98.8 %

Trips with >= 1 tip

13 / 13

Tests re-run here

VERDICT

TESTS 13/13

RE-RUN HERE

DECLARED SYNTHETIC

SCIKIT-LEARN

NO REAL TELEMETRY

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1. Objective and brief mapping

Brief: create an AI model to analyse driving behaviour and optimise energy usage with Python and scikit-learn, collect simulated driving data (speed, terrain, braking) and suggest driving tips from the predictions. I split it into four pieces: (1) a physics-based simulator of 2,000 EV trips at 1 Hz with eco / normal / aggressive driver archetypes on urban / highway / hilly roads, (2) 19 trip-level features that an OBD-II / GPS logger could also produce, (3) a Wh/km regressor (Ridge vs Random Forest vs Gradient Boosting, 5-fold CV + held-out test) and a driver-style classifier, (4) a recommender whose tips are counterfactual re-predictions of edited feature vectors.

All data is synthetic (README): "Every trip, speed trace, gradient profile and Wh/km value was generated by a simulator I wrote (src/data_generator.py). No real vehicle, driver or telemetry log was used. The numbers below describe how well the models learn the simulator, not how well they would work on a real fleet." python main.py runs the pipeline in 67.8 s (seed 42).

2. Model and architecture

Energy model per 1 Hz sample: $F_{\text{trac}} = m_{\text{eq}} a + m g C_{\text{rr}} \cos(\theta) + 0.5 \rho C_{\text{d}} A v^2 + m g \sin(\theta)$, $m = 1650 \text{ kg}$ + payload, $C_{\text{d}} A = 0.62 \text{ m}^2$, $C_{\text{rr}} = 0.010$ (x1.15 wet). Traction $P_{\text{batt}} = P_{\text{wheel}} / (\eta_{\text{motor}}(P) \eta_{\text{a}})$ with $\eta_{\text{motor}} = 0.94 - 0.18 \exp(-P/6 \text{ kW}) - 0.08 (P/110 \text{ kW})^2$; regen limited by a 1.5 m/s² envelope, 40 kW cap and a fade below 3 m/s, the rest lost as friction heat; 250 W base + HVAC auxiliaries. Drivers are proportional speed-tracking controllers with archetype-specific gain, acceleration limits, look-ahead and pedal jitter, perturbed per trip (log-normal, sigma 0.25).

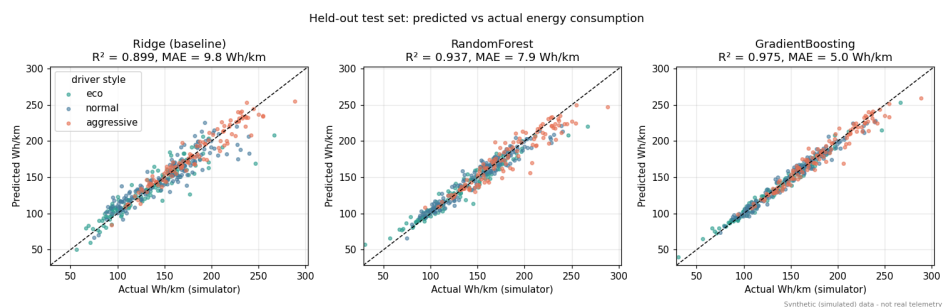


Fig. 1 - Predicted vs actual Wh/km on the 400 test trips for Ridge, Random Forest and GradientBoosting.

3. Method and implementation

- Split 1,600 train / 400 test, stratified by driver style; 5-fold CV on the training set, one evaluation on the untouched test set. GradientBoosting: 400 trees, depth 3, learning rate 0.05. Random Forest classifier on the 10 behaviour features only (no road / weather), so it must recognise how someone drives, not where.
- Permutation importance on the test set (delta R2 when shuffled): hvac_kw 0.577, net_climb_m_per_km 0.300, p95_speed_kmh 0.269, accel_std_ms2 0.111, mean_speed_kmh 0.057, speed_std_kmh 0.020, stops_per10km 0.015, mean_pos_accel_ms2 0.013.
- Recommender: six tips in TIP_LIBRARY, each with a firing rule and a coupled counterfactual edit (e.g. 'accelerate smoothly' sets harsh accel to 0, accel SD x0.75, speed SD x0.92); the model re-predicts, tips with no saving are dropped, the rest are ranked and a combined all-tips scenario is reported.

4. Results (results/metrics.json, model_comparison.csv)

Fleet: 2,000 trips, 41,259 km, 725 h. Mean Wh/km by style: eco 134.3 +/- 35.6 (n = 676), normal 150.5 +/- 35.7 (759), aggressive 178.7 +/- 38.4 (565); highway 181.2, urban 140.8, hilly 139.6. Harsh accel / brake events per 10 km: eco 0.5 / 0.1, normal 3.9 / 2.4, aggressive 6.7 / 5.7. Style classifier: accuracy 0.950, macro F1 0.951, per-class F1 eco 0.964 / normal 0.934 / aggressive 0.955; confusion matrix rows eco [133 2 0], normal [8 141 3], aggressive [0 7 106] - all errors between neighbouring classes.

Tips on the 400 test trips (trips fired / mean saving / mean Wh/km saved): climate control 202 / 10.0 % / 16.9; lower cruising speed 268 / 6.8 % / 12.4; accelerate smoothly 237 / 4.9 % / 8.0; brake gently 198 / 3.9 % / 6.5; cut stationary time 168 / 1.6 % / 2.6; remove weight 224 / 1.4 % / 2.2. Combined saving 19.9 % aggressive, 15.3 % normal, 8.3 % eco (mean 14.3 %); 98.75 % of trips received at least one tip.

Model	CV R2 (mean +/- SD)	CV MAE	Test R2	Test MAE	Test RMSE	Test MAPE
GradientBoosting	0.970 +/- 0.004	5.16	0.975	5.04 Wh/km	6.80	3.4 %
RandomForest (300 trees)	0.927 +/- 0.009	8.04	0.937	7.87 Wh/km	10.75	5.3 %
Ridge (baseline, scaled)	0.887 +/- 0.020	9.87	0.899	9.76 Wh/km	13.60	6.7 %

Wh/km (mean), 2,000 trips	eco	normal	aggressive	All styles
urban (n = 906)	128.4	136.9	159.6	140.8 +/- 36.9
highway (n = 619)	152.6	178.6	218.7	181.2 +/- 33.2
hilly (n = 475)	121.1	141.2	162.0	139.6 +/- 36.9

KEY TAKEAWAY A GradientBoosting model predicts Wh/km to 5 Wh/km on 400 held-out synthetic trips and a Random Forest recognises driving style at 95 %; counterfactual tips promise 8-20 % savings, but the data is my own simulator.

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Wh/km (mean), 2,000 trips	eco	normal	aggressive	All styles
All roads	134.3 +/- 35.6 (n = 676)	150.5 +/- 35.7 (n = 759)	178.7 +/- 38.4 (n = 565)	153.0 +/- 40.4

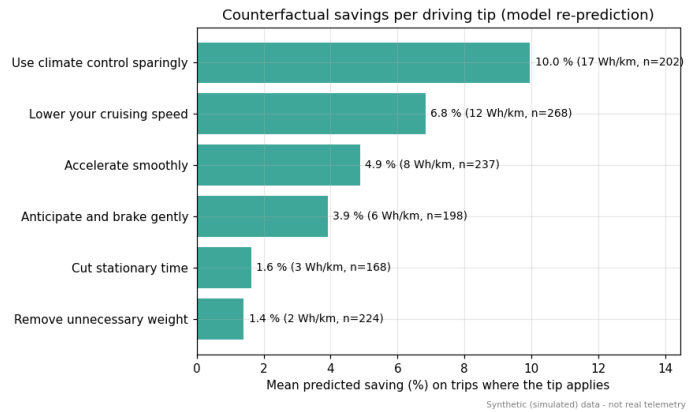


Fig. 2 - Predicted saving per tip and combined, by driver style, on the 400 held-out test trips.

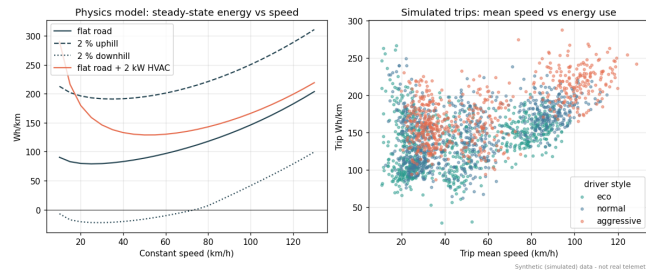


Fig. 3 - Steady-state Wh/km vs speed from the energy model, flat / uphill / downhill: the v^2 drag term that the tree models learn and the linear baseline cannot.

5. Verification

Re-run here with the fixture-aware runner (pytest not installed): **13 passed, 0 failed** in 17.8 s; the suite regenerates a dataset and retrains the models, so the numbers it checks are produced fresh in this session.

Test	Checks	Result
test_motor_efficiency_range, test_steady_state_energy_rises_with_speed, test Uphill_costs_more_than_flat, test_regen_never_exceeds_braking_power	physics model bounds and monotonicities	4 PASS
test_dataset_shape_and_energy_positive, test_aggressive_uses_more_energy_than_eco, test_generation_is_reproducible	positive energy and distance, no NaN features; aggressive > normal > eco; seed reproducibility	3 PASS
test_best_model_r2_on_holdout, test_tree_models_beat_linear_baseline, test_style_classifier_accuracy	R2 > 0.8 on held-out data; trees beat Ridge; classifier > 75 %	3 PASS
test_recommender_savings_non_negative, test_recommender_targets_aggressive_drivers, test_recommendation_markdown	every tip saving >= 0 and ranked, combined >= best single; aggressive > eco savings; markdown output	3 PASS

6. Data integrity and declared substitutions

- Synthetic data: the model is learning my simulator's equations back, which is why R2 is 0.97; real telemetry has sensor noise, GPS altitude error, battery-temperature effects and drivers who do not fit three archetypes - expect much lower R2 on real data.
- Counterfactuals are model extrapolations: editing one feature while holding others fixed can create combinations the model never saw; savings are indicative, not measured. HVAC dominates the tips because it is a large fixed load in the simulator; on a 44 degC afternoon that advice is not practical.
- Class labels are known because they are the generating archetype; real data would need clustering or annotation. README section 11 documents how to plug in OBD-II / CAN logs (PID 0x0D speed, V x I battery power, GPS grade) so that only the simulator changes.

7. Learnings

Two environmental factors (HVAC, climb) and two behavioural ones (cruising speed, acceleration smoothness) explain most of the variance, which matches the physics: HVAC is a fixed power spread over few kilometres at urban speeds and drag grows with v^2 . The linear baseline is already decent because Wh/km is nearly linear in HVAC and climb; the tree models earn their 8 % R2 by capturing the v^2 term and the interaction where harsh braking only hurts once it exceeds the regen envelope. A production coach would score tips by feasibility and comfort as well as Wh/km and give event-level rather than trip-level feedback.

KEY TAKEAWAY A GradientBoosting model predicts Wh/km to 5 Wh/km on 400 held-out synthetic trips and a Random Forest recognises driving style at 95 %; counterfactual tips promise 8-20 % savings, but the data is my own simulator.